Definition (Traced symmetric strict monoidal semicategory)
We say that a symmetric monoidal associative stacking semicategory $\langle \mathbf{C}, \otimes, \mathbf{1}, \sigma_s, \sigma_t \rangle$ is *traced* if it is equipped with a family of functions

$$\mathrm{Tr}^{Z}_{X,Y} : \mathrm{Hom}_{\mathbf{C}}(X \otimes Z; Y \otimes Z) \to \mathrm{Hom}_{\mathbf{C}}(X; Y),$$

satisfying the following axioms:

1. *Naturality in* X: For any morphisms $f : X \otimes Z \to Y \otimes Z$ and $g : X' \to X$, if Z has an identity id_Z, then

$$\mathrm{Tr}^{Z}_{X',Y}((g \otimes \mathrm{id}_Z) \fatsemi f) = g \fatsemi \mathrm{Tr}^{Z}_{X,Y}(f).$$

2. *Naturality in* Y: For any morphisms $f : X \otimes Z \to Y \otimes Z$ and $g : Y \to Y'$, if Z has an identity id_Z, then

$$\mathrm{Tr}^{Z}_{X,Y'}(f \fatsemi (g \otimes \mathrm{id}_Z)) = \mathrm{Tr}^{Z}_{X,Y}(f) \fatsemi g.$$

3. *Dinaturality in* Z: For any morphisms $f : X \otimes Z \to Y \otimes Z'$ and $g : Z' \to Z$, if id_X and id_Y exist, then

$$\mathrm{Tr}^{Z}_{X,Y}(f \fatsemi (\mathrm{id}_Y \otimes g)) = \mathrm{Tr}^{Z'}_{X,Y}((\mathrm{id}_X \otimes g) \fatsemi f).$$

4. *Vanishing I*: For any morphism $f : X \to Y$ in $\mathbf{C}$,

$$\mathrm{Tr}^{\mathbf{1}}_{X,Y}(f) = f.$$

5. *Vanishing II*: For any morphism $f : X \otimes Z \otimes U \to Y \otimes Z \otimes U$ in $\mathbf{C}$,

$$\mathrm{Tr}^{Z \otimes U}_{X,Y}(f) = \mathrm{Tr}^{Z}_{X,Y}\Big(\mathrm{Tr}^{U}_{X \otimes Z, Y \otimes Z}(f)\Big).$$

6. *Superposing*: For any morphisms $f : X \otimes Z \to Y \otimes Z$ and $g : V \to W$ in $\mathbf{C}$,

$$\mathrm{Tr}^{Z}_{V \otimes X, W \otimes Y}(g \otimes f) = g \otimes \mathrm{Tr}^{Z}_{X,Y}(f).$$

7. *Yanking*: For any object Z, if id_Z exists, then

$$(\pi_s \fatsemi \mathrm{Tr}^{Z}_{Z,Z})(\mathrm{id}_Z \otimes \mathrm{id}_Z) = \mathrm{id}_Z$$

and

$$(\pi_t \fatsemi \mathrm{Tr}^{Z}_{Z,Z})(\mathrm{id}_Z \otimes \mathrm{id}_Z) = \mathrm{id}_Z,$$

where $\pi \in S_2$ is the odd permutation corresponding to swapping the two copies of Z.